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Micromechanical properties of granite with insights into mineral interface mechanics

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ABSTRACT

Understanding the mechanical behavior of diagenetic mineral granules and interfaces in granite provides essential experimental references for constructing micromechanical models of granite. The micromechanical behavior of Yanshanian granite is investigated using scanning electron microscopy–energy dispersive spectroscopy (SEM-EDS) and nanoindentation tests. The results demonstrate transitional mechanical properties at mineral interfaces. The elastic modulus and hardness exhibit intermediate values between adjacent mineral phases. The higher plasticity indices at the interfaces suggest higher plastic deformation capacity of hard-phase minerals in these regions. Additionally, fracture toughness measurements of minerals and interfaces were obtained, with interfacial values ranging from 0.90 to 1.63 MPa·m$^{0.5}$. The analysis of mechanical property relationships shows a significant positive linear correlation between rock-scale elastic modulus and fracture toughness. However, this correlation is substantially lower at the mineral scale, demonstrating a scale effect in the relationship of different mechanical properties.

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1. Introduction

The depletion of fossil fuels, coupled with environmental issues, such as the greenhouse effect, rising regional acid rain, and nitrogen oxide pollution, has caused a shift toward environmentally friendly renewable energy sources [1]. Geothermal energy has become a research hotspot due to its widespread availability, cleanliness, and stability [2]. Granite is a common rock in hot dry rock (HDR) reservoirs [3]; thus, an in-depth study into granite's mechanical characteristics is required to guide long-term geothermal energy development [4].

Traditional tests like uniaxial and triaxial compression tests are typically conducted to assess the macromechanical properties of granite and obtain inputs for numerical simulations. However, HDR reservoirs are typically located at depths exceeding 3000 m, presenting significant challenges for conventional experimental methods. Core extraction at such depths is expensive and time-consuming, with low recovery rates of intact cores due to the release of high in-situ stresses, complicating the acquisition of reservoir rock mechanics data [5]. Nanoindentation testing is a promising solution to these challenges. Unlike conventional testing approaches, nanoindentation imposes minimal constraints on sample geometry and can be performed directly on irregular rock fragments, substantially reducing the time and costs associated with core extraction and specimen preparation [6]. Furthermore, this technique exhibits exceptional testing efficiency while providing nanoscale resolution for characterizing localized features, such as the mechanical properties of mineral phases and their interfaces [7]. Therefore, nanoindentation testing has been increasingly utilized in rock mechanics over the past decades and has recently been extended to analyzing extraterrestrial rock mechanics [8,9], offering an innovative approach to characterizing mechanical properties.

Many researchers have utilized nanoindentation to study the mechanical behavior of rocks, such as shale and coal. Researchers successfully linked diagenetic minerals to shale's micromechanical properties by combining nanoindentation with backscattered electron (BSE) diffraction analysis. Building on this foundation, Bennett et al. [10] investigated cross-scale physical models of shale, while Kumar et al. [11] analyzed the fundamental mechanical properties of specific minerals and their impact on macroscopic mechanical performance. Their findings indicated that a higher organic matter

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content reduced the rock's elastic properties. Mondol et al. [12] investigated the relationship between shale's elastic modulus and mineral composition. The results showed that a higher content of rigid matrices (such as quartz and feldspar) resulted in a higher elastic modulus, whereas a higher total organic carbon (TOC) content resulted in a lower elastic modulus. Liu and Ostadhassan [13] utilized field-emission scanning electron microscopy (SEM) with a nanoindentation test to examine the micromechanical properties of Bakken shale. They found that the elastic modulus was positively correlated with the hardness. Akono et al. [14] investigated how CO_2-driven chemical reactions influenced the mechanical properties of sandstone with particle sizes of 50 to 400 μm. They analyzed BSE images to visualize the evolution of sandstone minerals affected by CO_2 and successfully related the physical and chemical changes in rock minerals to their mechanical properties. Borodich et al. [15] prepared transparent coal thin sections and identified coal components using reflected light. They categorized coal into vitrinite (matrix), inertinite, and liptinite (inclusions) at the microscopic scale and employed nanoindentation test to assess the components' mechanical properties. Kossovich et al. [16] conducted nanoindentation tests on three coal types with different ranks. The study revealed that the vitrinite micro-components' mechanical properties varied with the coal rank, whereas the properties of inertinite showed minimal changes.

Studies on sedimentary rocks, such as shale and coal rock, have confirmed that the mechanical response of mineral components significantly affects the rocks' macroscopic mechanical behavior. However, few studies have been conducted on granites, especially on the mineral interfaces, which are key regions controlling rock deformation and fracture. This research gap constrains our in-depth understanding of the multiscale mechanical behavior of granites. Therefore, this paper adopts SEM, energy dispersive spectroscopy (EDS), and nanoindentation tests to systematically investigate the mechanical properties of granite minerals and their interfaces. The investigation elucidates the underlying mechanisms correlating mechanical properties with deformation and fracture behaviors and reveals a previously unreported scale effect in the relationships among mechanical properties of minerals. This study provides crucial experimental evidence for developing multiscale mechanical models that incorporate interfacial effects, advancing the theoretical framework for research on granite mechanics.

2. Experimental methodology of nanoindentation test

2.1. Preparation of granite sample

2.1.1. Geological background

The experimental samples were collected from the Guangdong-Hong Kong-Macao Greater Bay Area, where strata ranging from the Precambrian to the Quaternary are exposed. Since the Mesozoic, intense magmatic and tectonic activities have formed a 400-km-wide intrusive rock belt. Based on the intrusion ages, the intrusive rocks are classified into the Yanshanian, Indosinian, Hercynian, and Caledonian periods, with fault structures primarily developing under compressional stress environments from the Caledonian to the Yanshanian periods. The Yanshanian intrusive rocks cover the largest area, and the granite formations host abundant geothermal resources [17]. The rock samples were obtained from the Yanshanian granite formations.

2.1.2. Basic physical properties of the Yanshanian granite

The basic physical properties of the Yanshanian granite from the Greater Bay Area, such as elastic mechanical parameters and compressive strength, are presented in Table 1. The low water saturation indicates a low abundance of primary pores and natural fractures, suggesting a compact structure. The mineralogical composition was analyzed with a Smart Lab X-ray diffractometer (XRD) (Rigaku, Japan). As shown in Fig. 1, the primary diagenetic minerals in the Yanshanian granite are silicate minerals, such as quartz, feldspar, and biotite. In addition, trace amounts of carbonate minerals, such as dolomite, are present. The particle size distribution was analyzed utilizing a particle size analyzer (Malvern Panalytical, UK). The results (Fig. 2) display the frequency distribution and cumulative frequency curves of the particle sizes. Approximately 90% of the granite particles range from 30 to 600 μm, with less than 5% of particles smaller than 10 μm. This information provides a basis for determining the appropriate indentation depths for subsequent nanoindentation experiments.

2.1.3. Surface processing of sample for nanoindentation

The core samples were cut into cuboid slices with dimensions of 15 mm × 15 mm × 10 mm. The untreated surfaces of the samples exhibited local irregularities with roughness on the order of tens of micrometers. The surface roughness may introduce errors in nanoindentation test results. Therefore, surface grinding and polishing were necessary to minimize roughness and ensure accurate results. Mechanical and ion polishing were used to obtain a smooth and flat testing surface. The samples were placed on a Struers Tegramin-30 automatic polishing machine (Struers, Denmark), with the testing surface in contact with silicon carbide abrasive sheets. Sequential polishing was performed using abrasive sheets with grit sizes of P800, P1500, P2000, and P3500 for initial mechanical polishing. Subsequently, the polished samples were mounted on the sample stage of an IM4000 ion milling system (Hitachi, Japan) for fine ion polishing. Then, the samples underwent ultrasonic cleaning to remove surface debris and contaminants. The samples' surface roughness was evaluated using a Dimension Edge atomic force microscope (Bruker, USA). The sample schematic, surface preparation workflow, and results are shown in Fig. 3. The effectiveness of the surface treatment was quantified by calculating the root mean square roughness R_q [18], which is calculated by Eq. (1):

$$R_q = \sqrt{\frac{1}{T} \times \sum_{i=1}^{T} \left(J_i - \bar{J}\right)^2} \tag{1}$$

where J_i denotes the height of each data point; $\bar{J}$ the mean height; and T the number of data points. Fig. 3 indicates that the maximum absolute roughness of the testing surface is 2.3 nm. The calculated root mean square roughness R_q is 0.51 nm.

2.2. SEM-EDS testing

Advancements in microscopic detection technologies have made it feasible to study the deformation and failure characteristics of microscopic minerals. SEM-EDS is a technique capable of precisely determining mineral distributions by identifying elemental distributions and observing deformation features on mineral surfaces under external forces. As confirmed by the XRD analysis (Section 2.1.2), the Yanshanian granite was primarily composed of biotite (Bt), quartz (Qz), feldspar (Fs), and a small amount of carbonate minerals. The colors in the EDS maps denote the different elements. The minerals were determined using chemical formulas and localized by the SEM-EDS system. This study utilized a Scios 2 scanning electron microscope (Thermo Fisher Scientific, USA) in conjunction with a QUANTAX X-ray energy dispersive spectrometer (Bruker, USA). The scanning voltage and the current of SEM were 10 kV and 0.4 nA, respectively. The scanning voltage of EDS was 15 kV. Quartz is the easiest to identify because it contains only

Table 1
Basic properties of Yanshanian granites.

Number	Density (g/cm^3)	Water saturation (%)	Uniaxial compressive strength (MPa)	Elastic modulus (GPa)	Poisson's ratio
1	2.63	0.21	151.82	45.84	0.15
2	2.61	0.21	148.92	43.96	0.24
3	2.63	0.23	137.13	43.67	0.18
Average	2.62	0.22	145.96	44.49	0.19

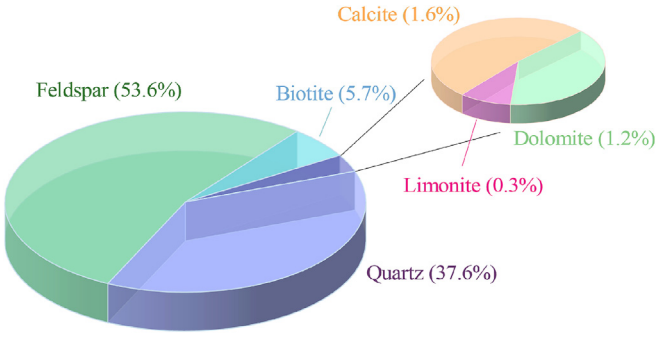


Fig. 1. Mineral composition of the Yanshanian granite.

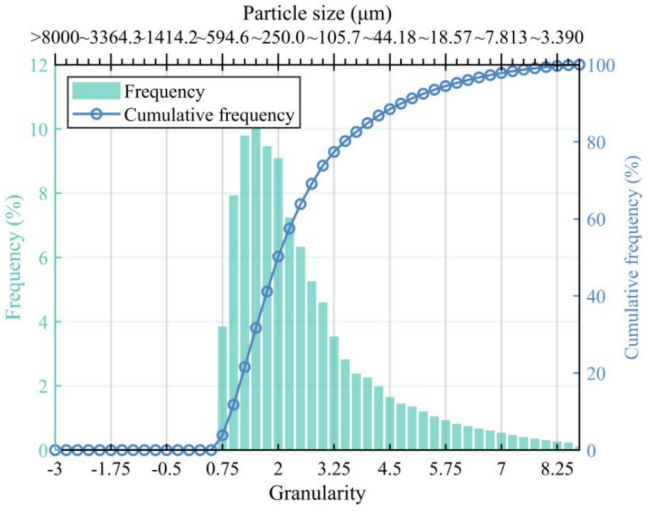


Fig. 2. Particle size distribution of the Yanshanian granite.

silicon and oxygen (SiO_2). In contrast, feldspar is rich in aluminum and sodium or potassium ($(K, Na)AlSi_3O_8$), which allows for distinguishing the minerals. Biotite contains magnesium or iron ($K(Mg, Fe)_3(AlSi_3O_{10})(F, OH)_2$), elements that are absent in quartz and feldspar. These distinguishing features enable the clear differentiation of the three primary target minerals and the precise identification of mineral boundaries (Fig. 4).

2.3. Nanoindentation procedure

The nanoindentation tests were performed using a Hysitron TI 990 (Bruker, USA) equipped with a Berkovich indenter. The mechanical parameters obtained through nanoindentation are highly dependent on the indentation depth. Previous studies have shown that varying peak loads applied by the indenter can result in significant fluctuations in the measured mechanical parameters [19]. Since a well-defined univariate functional relationship exists between the indentation depth and the applied load, it is crucial to select the appropriate load for the tests. The indentation depth must significantly exceed the surface roughness index and be much smaller than the target phase dimension in the sample to obtain accurate mechanical information for single-phase media. The indentation depth parameters were designed based on the submicron mineral distribution (Section 2.1.2) and surface roughness index (Section 2.1.3). The root mean square roughness R_q and approximately 93% of the mineral size's range ([30 μm, 600 μm]) were used as benchmarks. The lower limit of the indentation depth was 20 times the R_q, and the upper limit was one-tenth of the lower bound of the mineral size's range. Pre-tests were conducted on each phase medium to determine the optimal indentation parameters. Ultimately, the peak load for the indenter was 200 mN. A grid indentation strategy was employed to obtain statistically significant measurements of the mechanical parameters for each single-phase medium because of the minerals' inherent anisotropy and heterogeneity. For two-phase interfaces, multiple indentations were performed along the interface, with precise positioning guided by the results of the SEM-EDS analysis. The spacing between indentation points was at least three times the indentation depth to minimize the influence of plastic deformation from adjacent indentations. Based on the maximum depth obtained from the 200 mN pre-test, the spacing between indentation points was 30 μm. The layout of the indentation points and their relative positions are illustrated in Fig. 5.

The loading and unloading protocol used in the experiment is illustrated in Fig. 6a. The load was applied at 5 mN/s during the loading stage, followed by a 30-second hold at the peak load. Subsequently, the load was unloaded at 5 mN/s until the indenter detached from the material surface. The ideal load–displacement curve is shown in Fig. 6b, The shaded part of the figure corresponds to the calculation of fracture toughness, as detailed in Appendix B. During the loading phase, the curve contains information on the material's elastic–plastic deformation caused by the indenter. Additionally, "pop-in" phenomena may appear on the curve during this phase. They are primarily associated with deformation mechanisms, such as crack formation, propagation, or material chipping. Deformation continues under a constant load in the holding stage due to the material's plasticity, increasing the indentation depth. This stage is typically used to obtain material creep information and minimize displacement measurement errors caused by thermal drift. During the unloading phase, the material surface exhibits recovery as the indented is removed. This recoverable deformation corresponds to the material's elastic deformation. The unloading curve captures the material's elastic behavior. The elastic modulus (E) and hardness (H) can be determined from this phase using the Oliver and Pharr (OP) method [20]. This method defines the reduced modulus E_r according to Hertzian contact theory (Eq. (2)):

$$\frac{1}{E_r} = \frac{1-\nu^2}{E} + \frac{1-\nu_i^2}{E_i} \tag{2}$$

where ν denotes the Poisson's ratio. It was 0.2 based on the macroscopic sample's experimental values (Table 1). This assumption is reasonable because variations in ν within its typical range have minimal impact on the calculated value of the reduced modulus E_r (Supplementary material). E denotes the elastic modulus; E_i the elastic modulus of the indenter (it is typically 1141 GPa); and ν_i the Poisson's ratio of the indenter (it is typically 0.07).

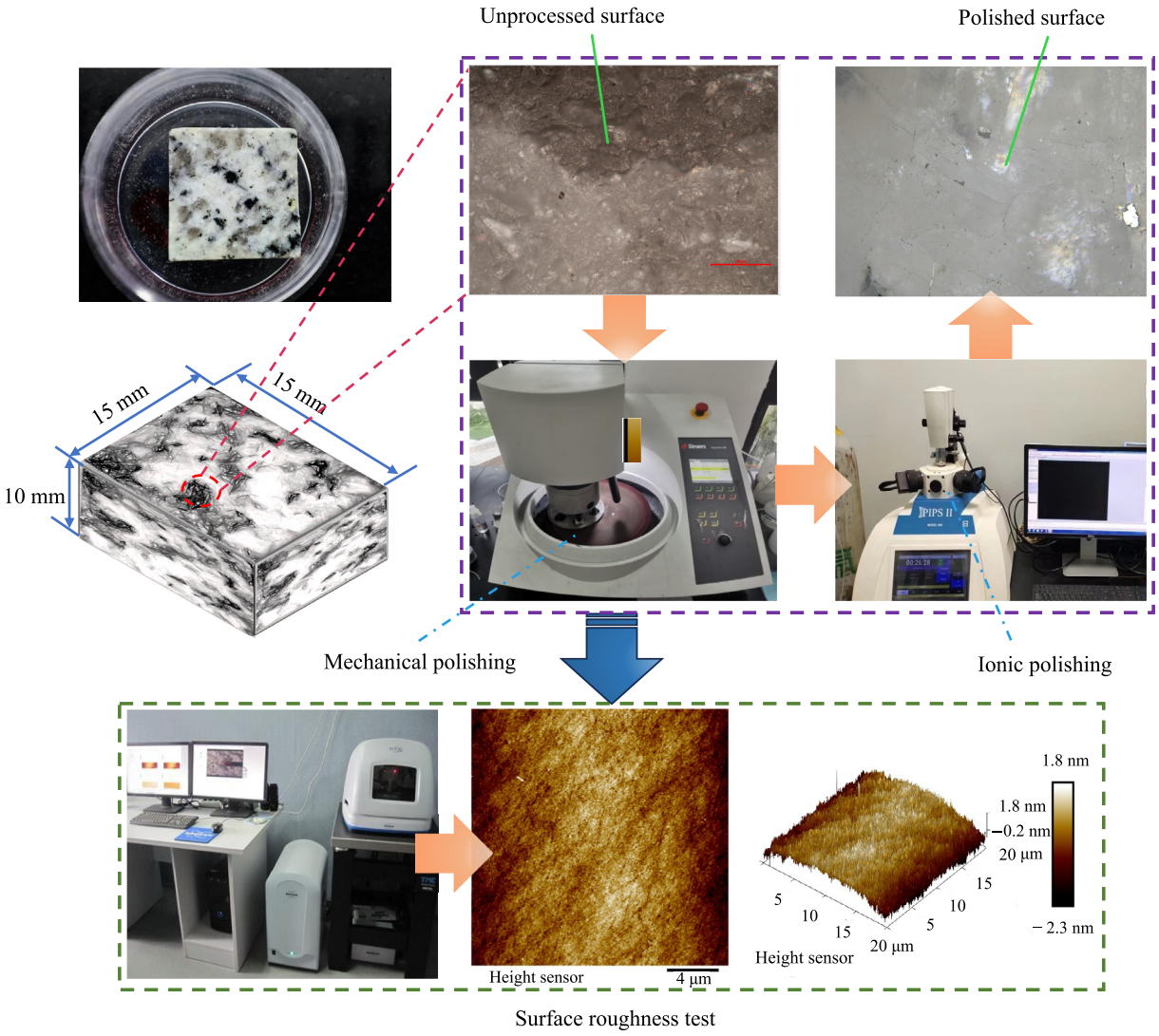


Fig. 3. Surface processing and roughness test of granite samples.

Based on contact mechanics theory, E_r is given by Eq. (3):

$$E_r = \frac{\sqrt{\pi}}{2\alpha}\frac{S}{\sqrt{A_c}} \tag{3}$$

where S is the contact stiffness; α a geometry-dependent correction factor for the Berkovich indenter, which is typically 1.034; and A_c the projected contact area.

A_c depends on the contact depth h_c, which is calculated by Eqs. (4) and (5):

$$h_c = h_m - \varepsilon\frac{P_m}{S} \tag{4}$$

$$A_c = 24.56h_c^2 \tag{5}$$

where h_m is the maximum indentation depth (Fig. 6b); ε typically 0.75 for a Berkovich indenter; and P_m the maximum applied load.

The above equations are solved by calculating S, which is defined in Eq. (6):

$$S = \left.\frac{dP}{dh}\right|_{h=h_m} \tag{6}$$

According to research by Oliver and Pharr, the unloading segment can be well approximated using a power-law relationship. This fit is typically applied from the beginning of unloading to approximately 50% of P_m. The curve is calculated using Eq. (7):

$$P_{ul} = \beta(h - h_f)^n \tag{7}$$

where P_{ul} is the load during the unloading phase; h_f the residual depth at zero load (Fig. 6b), and β and n the fitting parameters. Accordingly, the contact stiffness S can be calculated by Eq. (8):

$$S = \beta n(h_m - h_f)^{n-1} \tag{8}$$

Thus, E is calculated by solving Eqs. (1)–(8). H is defined by Eq. (9):

$$H = \frac{P_m}{A_c} \tag{9}$$

In addition to calculating E and H, the fracture toughness (K_{IC}) of the granite's microscopic compositions is determined. The calculation method is detailed in Supplementary material. In summary, the mechanical properties of the principal diagenetic minerals were obtained, along with those of the interfaces between quartz and feldspar (Qz-Fs), quartz and biotite (Qz-Bt), and feldspar and biotite (Fs-Bt) in the Yanshanian granite from the Greater Bay Area. These results provide an insight into the mechanical behavior of the granite at the microscale.

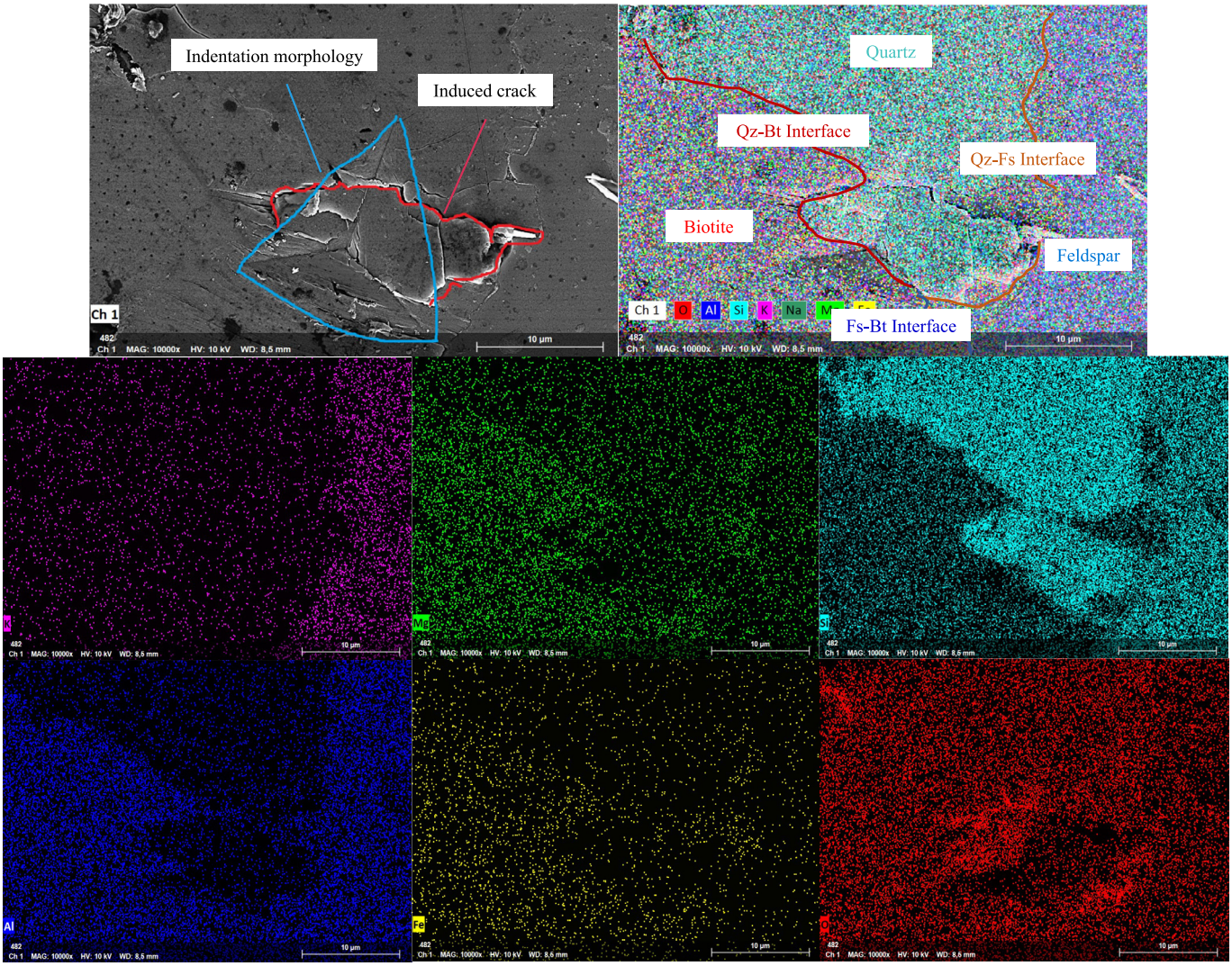


Fig. 4. Schematic of mineral surface morphology at the micro-scale and determination of sample minerals with mutual interfaces by SEM-EDS.

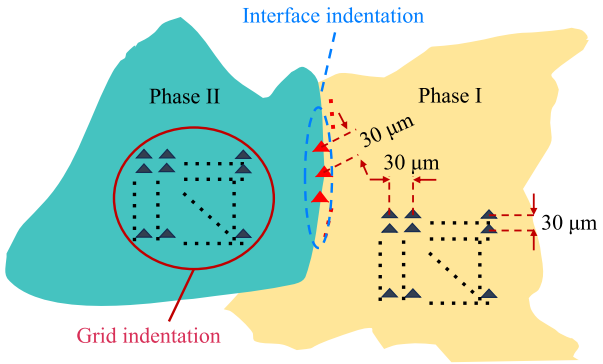
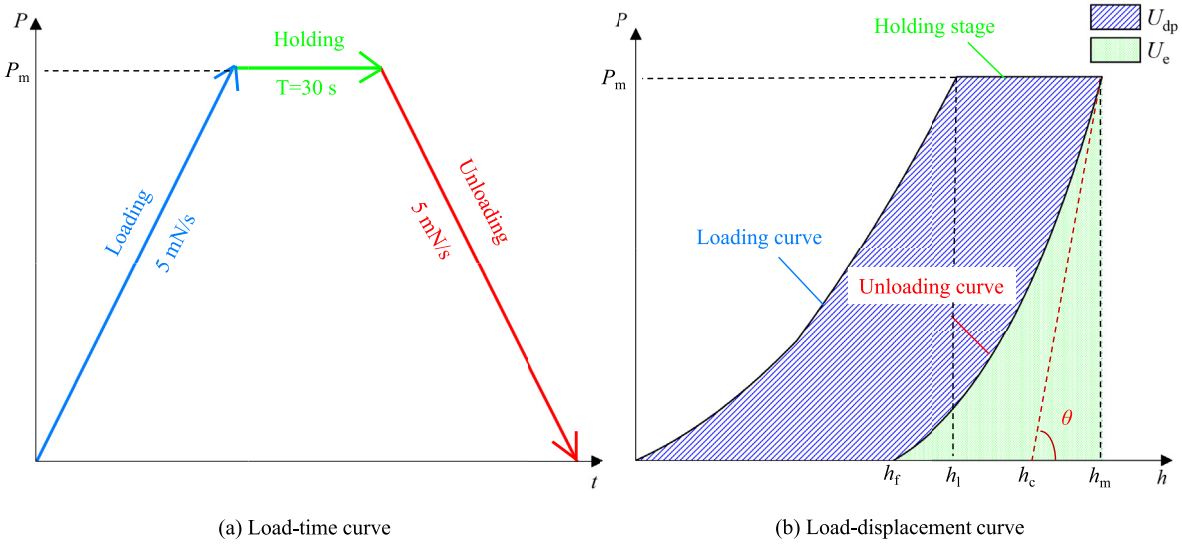


Fig. 5. Schematic of grid indentation on a single mineral and at the minerals' interfaces.

3. Results and analysis

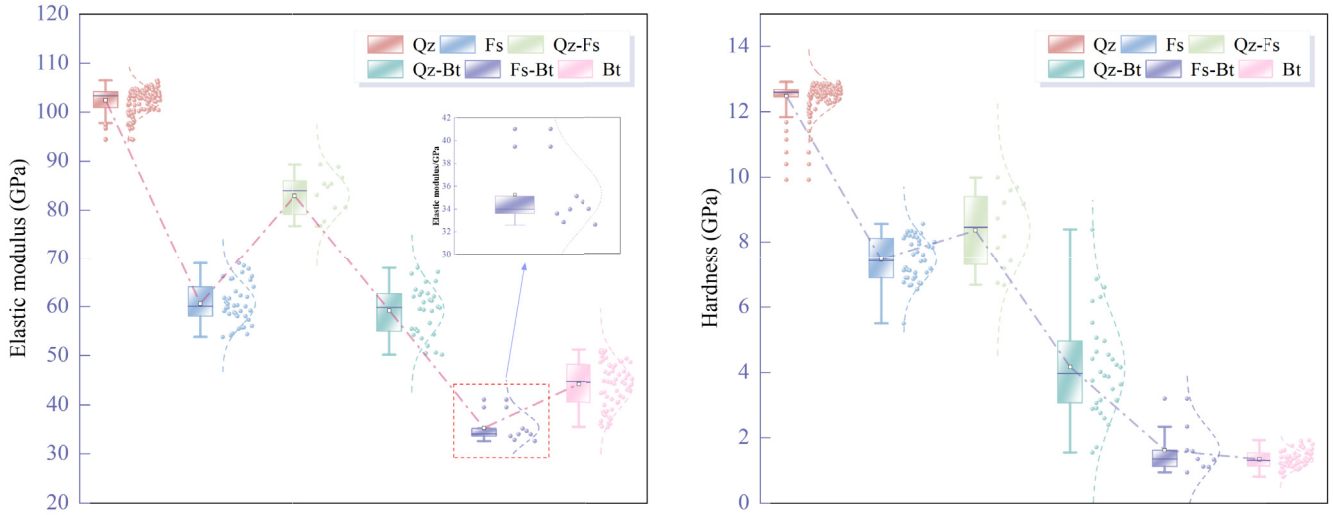
3.1. Elastic modulus and hardness of minerals and their interfaces

The elastic modulus characteristics of quartz, feldspar, biotite, and the quartz and feldspar (Qz-Fs), quartz and biotite (Qz-Bt), and feldspar and biotite (Fs-Bt) interfaces are presented in Fig. 7a. The hardness characteristics are shown in Fig. 7b. The relative magnitudes of the *E* and *H* exhibit consistent trends for all minerals and interfaces. For single-phase minerals, quartz shows the highest *E* and *H*, followed by feldspar, whereas biotite has the lowest values for both properties. Mineral interfaces generally show weaker mechanical properties than the strongest single-phase mineral at the interface, reflecting the mechanical weakening effect of interfaces. *E* and *H* decrease in the order of Qz-Fs, Qz-Bt, and Fs-Bt. This suggests that the boundaries of the mechanically stronger minerals have superior mechanical properties. The indentation experimental results of Liu et al. [21] support this observation, although they contrast with the conclusions drawn by Tang et al. [22] using atomic force microscopy (AFM) experiments. The single-phase mineral results from both studies are largely consistent, whereas significant discrepancies occur in the interface parameters, indicating that non-contact experimental techniques might be more sensitive to subtle perturbations at the microscale. This result highlights a critical direction for future research to better understand the mechanical properties of interfaces using different experimental approaches. The mechanical parameters of minerals and interfaces are staggered, with quartz displaying the highest mechanical performance. The Qz-Fs interface has stronger mechanical properties than feldspar and can be regarded as a strong interface, whereas the Fs-Bt interface shows



(a) Load-time curve (b) Load-displacement curve

Fig. 6. Schematic of loading, holding, and unloading for a single indentation point.



(a) Elastic modulus (b) Hardness

Fig. 7. Mechanical properties of minerals and interfaces.

the weakest mechanical performance, suggesting the biotite-containing interfaces are weak.

A statistical evaluation was conducted to analyze the dispersion of minerals and interfaces' mechanical properties. The nondimensional coefficient of variation (CV), denoted as φ, was used to quantify the dispersion of E and H of the minerals and interfaces. CV is defined by Eq. (10):

$$\varphi = \sqrt{\frac{\sum_{i=1}^{N}(x_i-\mu)^2}{\mu^2(N-1)}} \times 100\% \tag{10}$$

where μ is the mean value of the measured mechanical parameters; x_i the measured data at each indentation point; N the total number of data points; and φ the proportional to the degree of data variability.

Fig. 8a–c illustrates the elastic modulus distributions for the single-phase minerals (quartz, feldspar, and biotite). The ranges are 94.30–106.40 GPa, 53.75–68.86 GPa, and 35.45–51.21 GPa, respectively. The corresponding φ values are 2.54%, 7.03%, and 10.78%. Fig. 8d–f presents the hardness distributions, where the ranges for quartz, feldspar, and biotite are 9.91–12.91 GPa, 5.49–8.56 GPa, and 0.80–1.92 GPa, with φ values of 3.53%, 9.18%, and 19.88%, respectively. These results indicate that the degree of variability in E and H follows the order biotite>feldspar>quartz. Quartz has the smallest φ value, suggesting minimal variability in its mechanical properties. In contrast, biotite exhibits broader histograms, wider peak spacing, and larger φ values, indicating higher variability in its mechanical properties. This result suggests that quartz possesses a relatively homogeneous structure, whereas biotite exhibits pronounced heterogeneity. These observations are consistent with the inherent structural characteristics of the minerals. Fig. 9a–f depicts the mechanical property distributions of the mineral interfaces. All exhibit relatively dispersed distributions. Notably, the φ value for the E of the Fs-Bt interface is 8.43%, slightly lower than the 8.67% of the Qz-Bt interface. This finding suggests

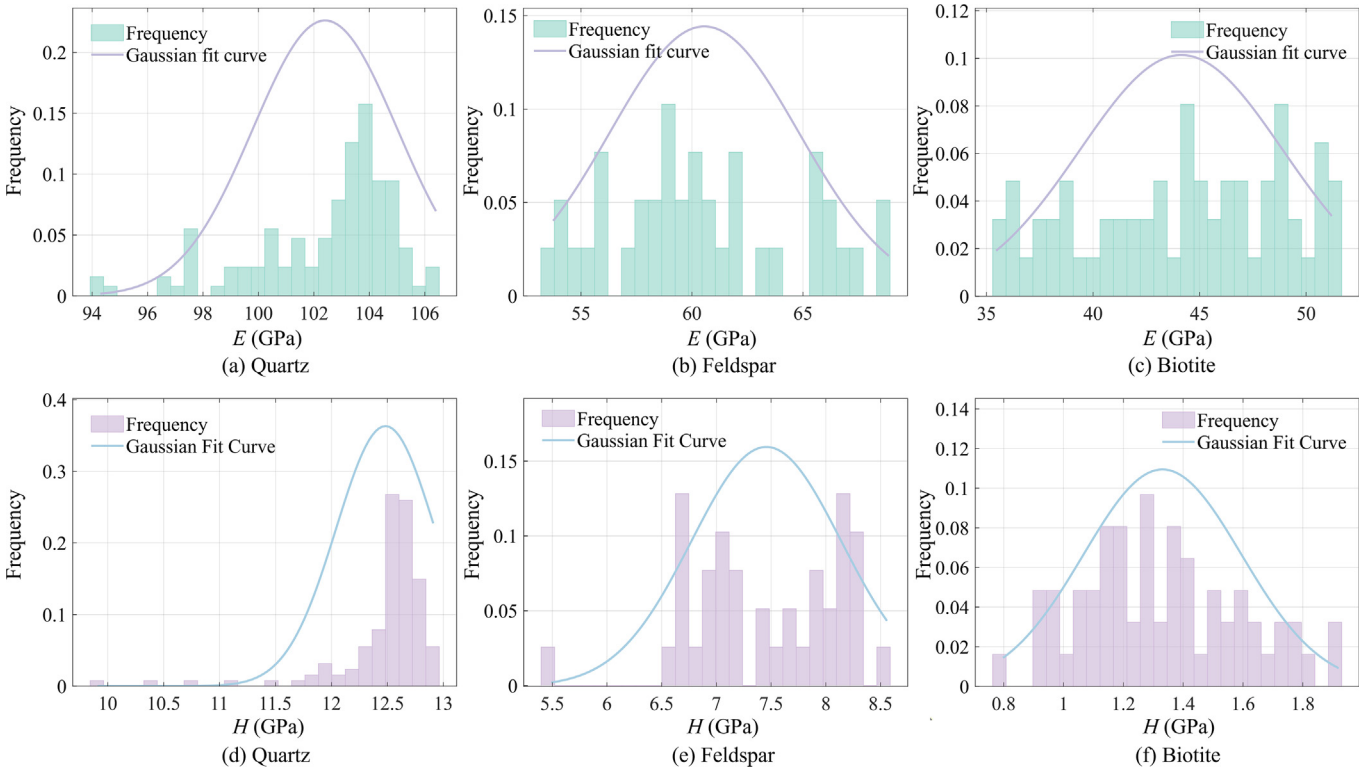


Fig. 8. Distribution of mechanical properties of single-phase minerals.

that the variability in interface mechanical properties may depend largely on the mechanical differences between the constituent phases of the interface. The hardness variability is significantly higher for interfaces than single-phase minerals, with φ values for biotite-related interfaces of 40%. This result highlights the pronounced heterogeneity in the mechanical behavior of these interfaces. Additional data are provided in Table 2. The mean E values of the single-phase minerals and interfaces are higher than the uni-

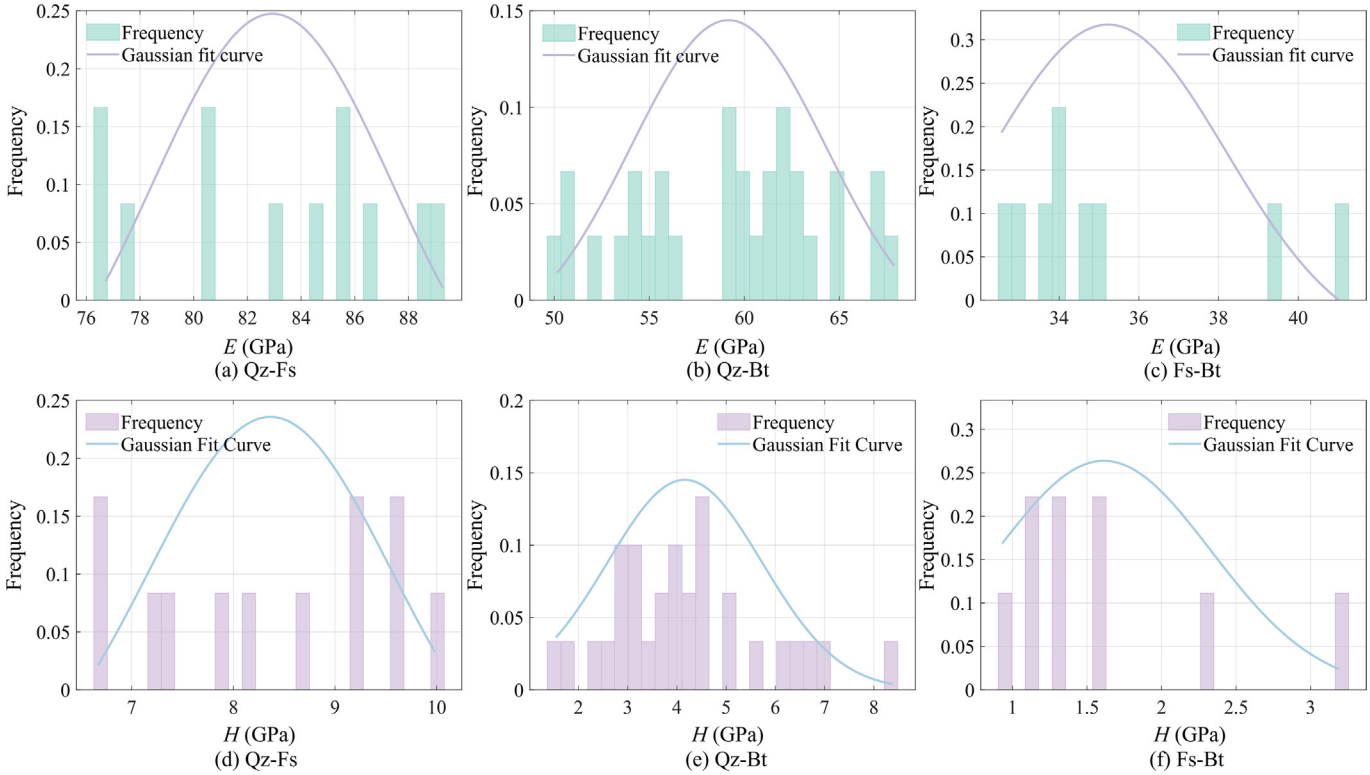


Fig. 9. Distribution of mechanical properties of interfaces.

axial elastic modulus of the macroscopic aggregate (Table 1), reflecting the size effect in the mechanical properties of granite.

3.2. Microscopic elasto-plastic deformation and plasticity index

The load–displacement curve serves as the foundation for studying the deformation characteristics of minerals and their interfaces. Five curves were plotted for each single-phase mineral and interface, as shown in Fig. 10a–f. Quartz exhibits the strongest and most stable deformation resistance among the single-phase minerals, with a maximum displacement of approximately 1100 nm under a 200 mN load. Its deformation paths remain nearly consistent across all curves. Biotite's load–displacement curves are significantly different from those of quartz and feldspar. A highly pronounced "pop-in" phenomenon occurs during the mid-loading phase, which is characterized by abrupt increases in displacement. Such high-frequency "pop-in" events are considered a distinctive feature of layered minerals [23]. Biotite exhibits the lowest resistance to deformation, with highly dispersed displacement paths and a maximum displacement of approximately 3000 nm under a 200 mN load—nearly three times that of quartz. This difference in deformation behavior is attributed to the structural characteristics of the mineral grains. At the microscale, quartz and feldspar grains have dense structures with almost no micropores or primary cracks (Fig. 11a and b). In contrast, biotite grains show observable microcracks (Fig. 11c), and these structural defects increase the sensitivity of biotite to stress-induced deformation. Additionally, a plastic indentation is observed along one edge of the biotite at the 10 μm scale, with ripple-like cleavage forming fiber-like strips around the indentation. These strips are prone to peeling off in lamellar shapes at the indentation edges (Fig. 12d), suggesting that biotite's structure consists of layered stacks of ripple-like cleavages formed by parallel grain alignment. This structure explains biotite's lower mechanical strength and deformation resistance.

The Qz-Fs interface exhibits a curved profile analogous to the mechanical behavior of quartz and feldspar, demonstrating robust deformation resistance. Conversely, interfaces associated with biotite, such as Qz-Bt and Fs-Bt, exhibit pronounced "pop-in" phenomena under stress conditions. Notably, the Fs-Bt interface has greater displacement than biotite-dominated regions, indicating a lower boundary stiffness and deformation resistance. These findings imply that biotite-enriched interfaces are more susceptible to deformation and fracturing under equivalent loading conditions, with crack propagation predominantly occurring along intergranular pathways [24]. The results in Section 3.1 indicate that the hardness variability is significantly greater for the interfaces than for single-phase minerals. This disparity in hardness variability reflects the more complex deformation behavior of interfaces than homogeneous single-phase minerals. Fig. 12a–f illustrates representative indentation morphologies at three types of boundaries, compared with those observed in single-phase minerals. Fig. 12a depicts the indentation morphology at the Qz-Bt boundary, where distinct stratification is evident. The cross-section of the indentation appears smooth on the quartz side, whereas flaky delamination occurs on the biotite side. A similar trend is observed in Fig. 12b, highlighting the complex deformation mechanisms at the interfaces. In contrast to the uniformity of indentation morphologies in single-phase minerals (Fig. 12e and f), the interface indentation morphologies display an anisotropic pattern characterized by elongation in a specific direction. Moreover, material accumulation called "pile-up" is evident on the weaker mineral phase side (Fig. 12c), which is attributed to the localized plastic deformation of the material. This behavior underscores the inherent complexity of interfacial deformation compared to homogeneous mineral phases.

Plastic behavior profoundly influences the deformation mechanisms of minerals and their interfaces, whereas the micro-scale elastoplastic behavior of rock materials governs their macro-scale failure modes (brittle or ductile). Therefore, the elastoplastic behaviors of minerals and interfaces were quantified to improve the understanding of micro-scale deformation in granite. The elastoplastic properties of minerals and interfaces can be derived from their indentation curves. The unloading segment of the curve corresponds to the elastic recovery phase (as discussed in Section 2.3). Consequently, the ratio of the h_f to the h_m reflects the extent of irreversible plastic deformation. Based on this, a dimensionless plasticity index P-value is defined by Eq. (11):

$$P = \frac{h_f}{h_m} \tag{11}$$

The theoretical range of the P-value is $0<P<1$. A P-value of 0 (h_f=0) indicates complete recovery of indentation displacement during unloading, which corresponds to elastic deformation. Con-

Table 2
Mechanical properties of microscopic minerals and interfaces.

Reference	Type	E_{mean} (GPa)	H_{mean} (GPa)	σ_E (GPa)	σ_H (GPa)	φ_E (%)	φ_H (%)
This study	Quartz	102.40	12.48	2.60	0.44	2.54	3.53
	Feldspar	60.56	7.46	4.26	0.68	7.03	9.18
	Biotite	44.14	1.33	4.76	0.26	10.78	19.88
	Qz-Fs	82.93	8.36	4.46	1.19	5.38	14.21
	Qz-Bt	59.16	4.16	5.13	1.57	8.67	37.76
	Fs-Bt	35.22	1.61	2.97	0.72	8.43	44.66
Tang et al. [22]	Quartz	96.81		1.01			
	Feldspar	69.50					
	Biotite	49.85		10.07			
	Qz-Fs	26.90					
	Qz-Bt	25.15					
	Fs-Bt	21.39					
Liu et al. [21]	Quartz	106.30	14.50	2.90	0.90		
	Orthoclase	100.60	11.40	1.20	0.40		
	Biotite	45.20	1.80	6.30	0.50		
	Plagioclase	67.20	3.20	6.40	0.50		
	Qz-Bt	56.00	2.80	6.70	0.80		
	Qz-Pl	71.40	8.20	8.90	1.50		
	Qz-Ol	91.60	12.10	9.70	1.70		
	Ol-Pl	57.40	2.50	7.90	0.40		
	Ol-Bt	56.40	3.60	5.20	1.10		

Note: Qz-Pl represents the interface of quartz and plagioclase; Qz-Ol the interface of quartz and orthoclase; Ol-Pl the interface of orthoclase and plagioclase; Ol-Bt the interface of orthoclase and biotite.

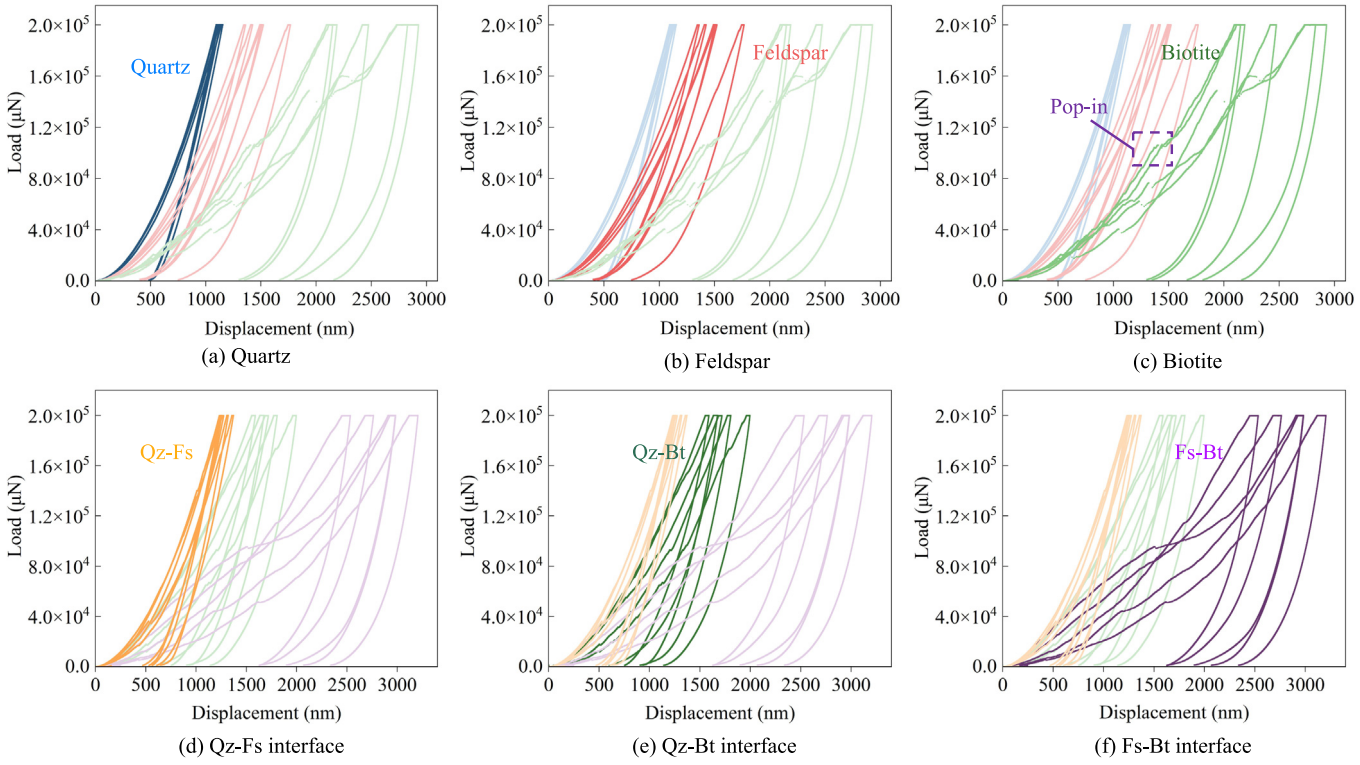


Fig. 10. Load-displacement curves for single-phase minerals and interfaces.

versely, a *P*-value of 1 (h_f=h_m) represents no recovery of deformation during unloading, indicating plastic behavior. The statistical distribution of the *P*-values for minerals and interfaces derived from Eq. (11) is illustrated in Fig. 13. For single-phase minerals, quartz and feldspar exhibit relatively low *P*-values, indicative of predominantly elastic deformation characteristics (Fig. 12e and f). In contrast, biotite has the highest *P*-value, reflecting its strong plasticity, as corroborated by its complex indentation morphology (Fig. 12d). The plasticity of low-*P*-value minerals is higher at the interfaces. In contrast, biotite interfaces generally exhibit higher *P*-values, suggesting they are prone to complex elastoplastic deformation (Fig. 12a and b). This observation aligns with the increased plasticity and deformation heterogeneity observed at biotite interfaces.

3.3. Microscopic fracture mechanisms and fracture toughness

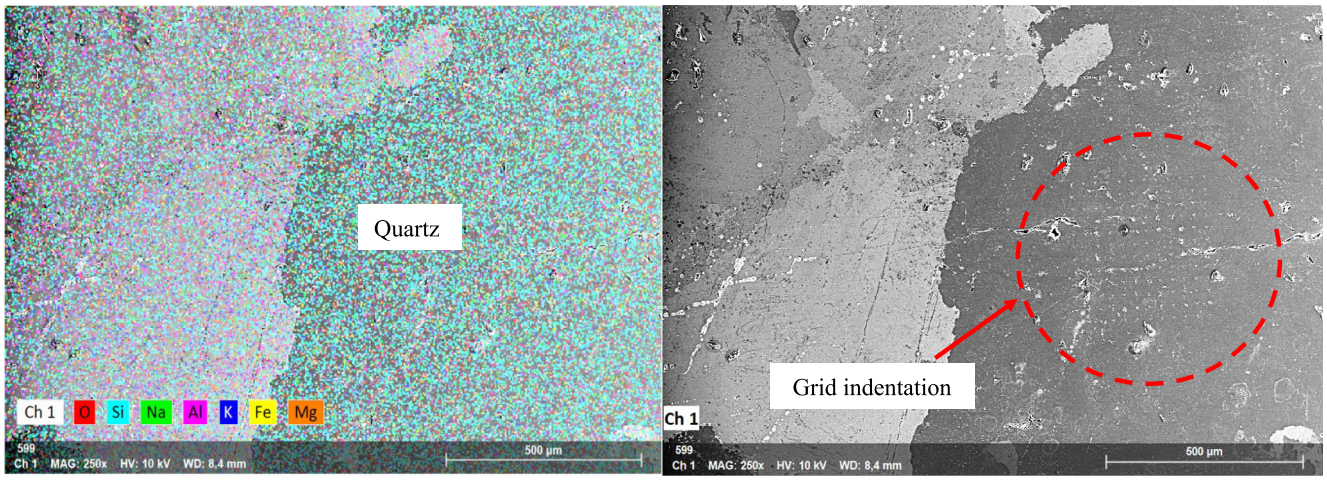
Fracture toughness K_{IC} is crucial for characterizing the resistance to crack propagation of materials. It is important in fracture network modeling and hydraulic fracturing design in oil and gas extraction, as well as geothermal engineering applications. The statistical results of K_{IC} for granite minerals and their interfaces are presented in Fig. 14. The K_{IC} of the minerals and interfaces in granite ranged from 0.86 to 2.02 MPa·m$^{0.5}$, with interfacial K_{IC} values from 0.90 to 1.63 MPa·m$^{0.5}$. Notably, feldspar has slightly lower K_{IC} than the Qz-Fs interface, making it more prone to intragranular cracks. Similar characteristics were reported by Liu et al. [21] and Fujii et al. [25]. Table 3 presents findings from various studies on the K_{IC} of granite, including macro-scale Brazilian disk splitting experiments [26], notched three-point bending tests [27], micro-scale methods, such as nanoindentation experiments [21,28], and micro-cantilever bending tests [29]. The fracture toughness values obtained for minerals and interfaces in this study are more consistent with those derived from traditional macro-scale experiments. The K_{IC} of hard minerals, such as quartz and feldspar, is almost identical to that of the macroscopic granite. Given their substantial proportion in granite (Fig. 1), quartz and feldspar play a dominant role in granite's resistance to crack initiation.

Previous studies on metallic and ceramic materials generally calculated fracture toughness using Eq. (12) [30]:

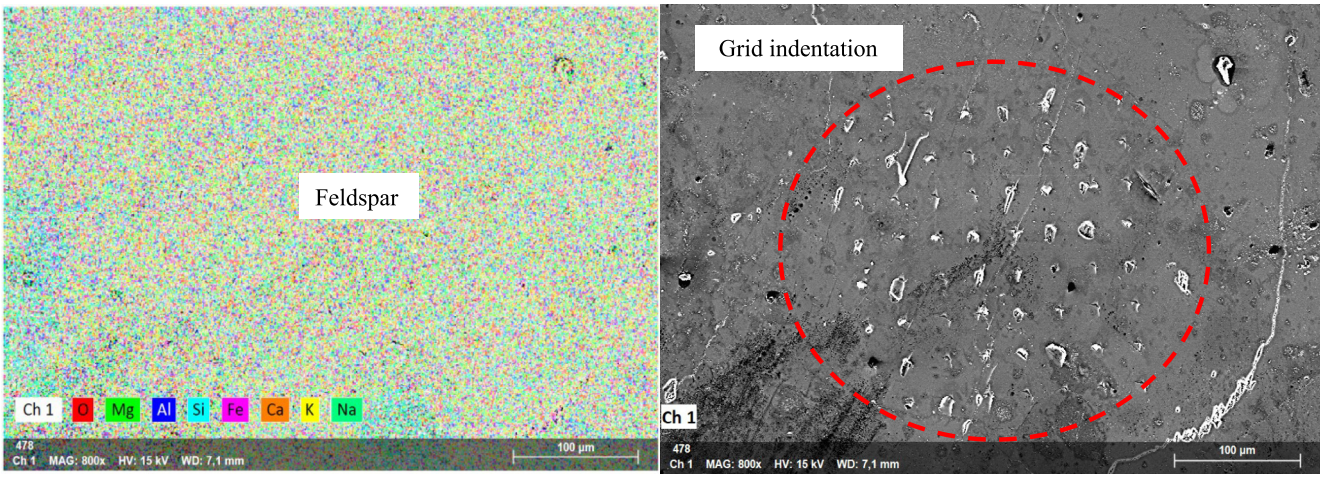
$$K_{IC} = \gamma \left(\frac{E}{H}\right)^{\frac{1}{2}} \left(\frac{P_{max}}{c^{\frac{3}{2}}}\right) \tag{12}$$

where γ is the indenter geometry constant; and c the average length of a crack extending from the indentation tip.

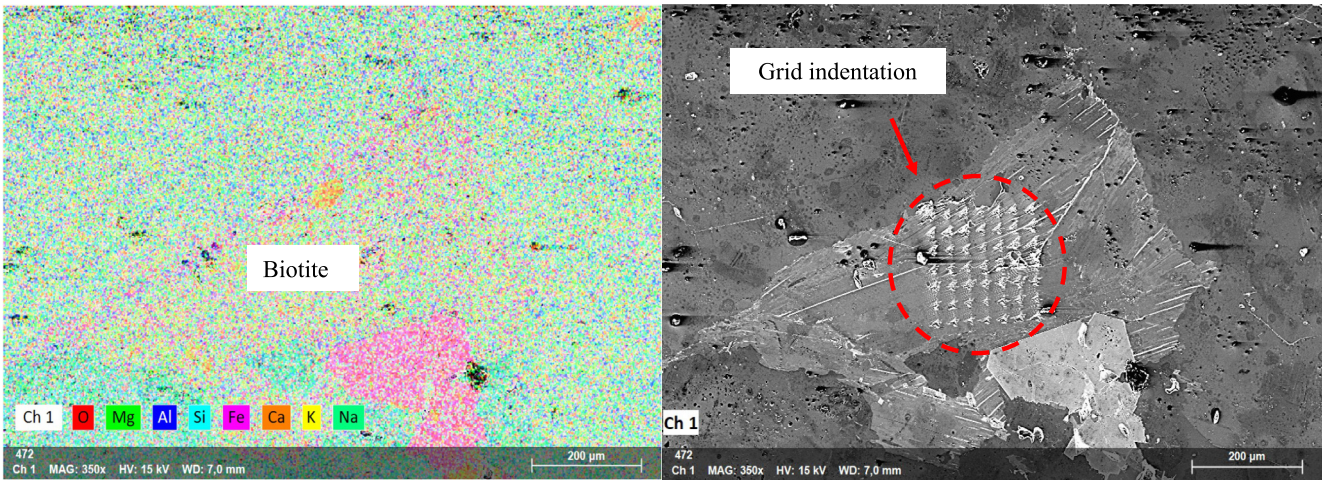
This model has been widely used to measure fracture toughness due to its simplicity and incorporation of crack length. However, this method is not applicable to the heterogeneous materials in this study. Fig. 15a–f illustrates the typical fracture morphologies of single-phase minerals and mineral interfaces. Only quartz–with its relatively high homogeneity–developed radial cracks extending from the indentation tips (Fig. 15a). Feldspar also exhibited distinct radial cracks with angular deviations from the indentation tips during propagation, indicating a fracture pattern controlled by internal jointing. In contrast, biotite appears to be less prone to the formation of tip cracks. A notable difference from the other two minerals is the development of numerous microcracks in the cleavage directions around the indentation. These microcracks exhibit plastic characteristics (Section 3.2), explaining the phenomenon observed in some experimental studies where cracks in granite propagated along biotite grain boundaries rather than through the biotite [21,25]. Despite biotite's relatively low K_{IC}, which makes it more prone to cracking under the same stress conditions (Fig. 15a–c), its high plasticity significantly affects crack propagation [31]. Several factors influence the behavior when a crack reaches a biotite grain boundary. First, if the crack attempts to penetrate through the biotite grain boundary, more fracture energy is required to overcome the plastic energy dissipation of



(a) Quartz

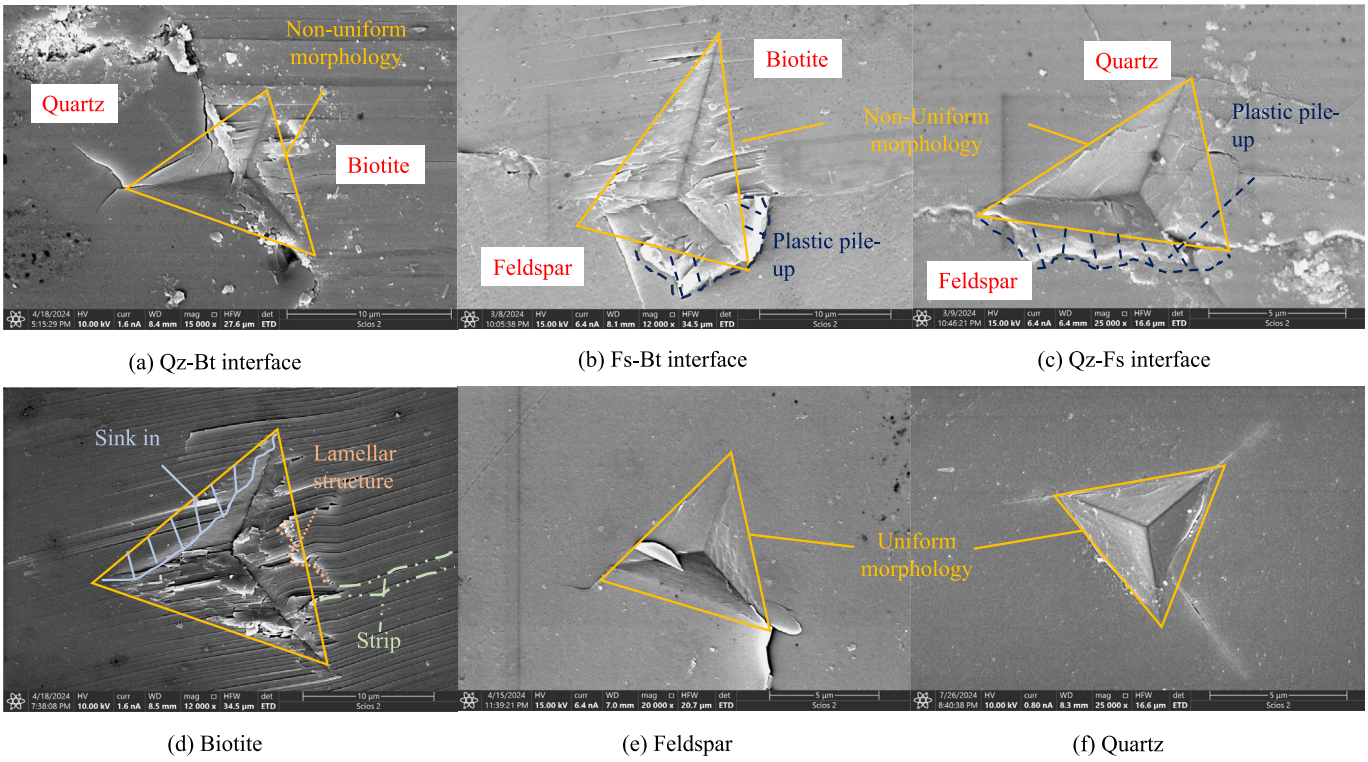


(b) Feldspar



(c) Biotite

Fig. 11. Surface morphology of minerals.



(a) Qz-Bt interface (b) Fs-Bt interface (c) Qz-Fs interface

(d) Biotite (e) Feldspar (f) Quartz

Fig. 12. Deformation morphology of single-phase minerals and interfaces.

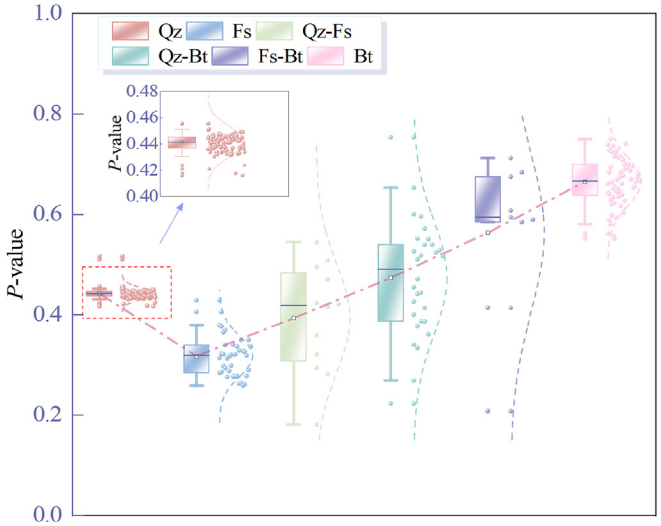


Fig. 13. *P*-value of single-phase minerals and interfaces.

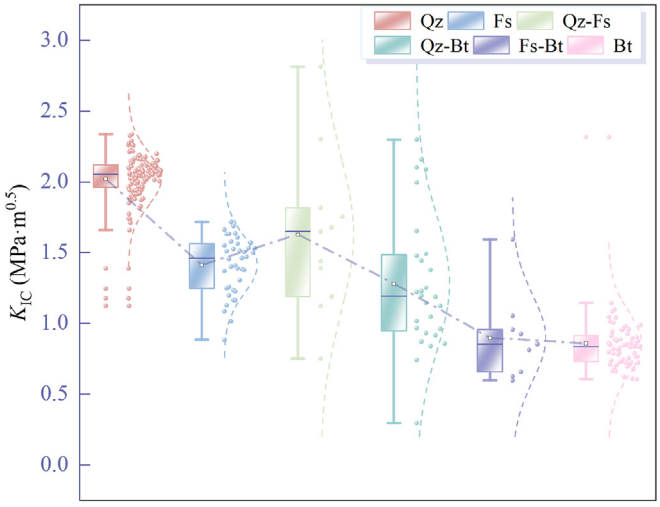


Fig. 14. Fracture toughness K_{IC} of single-phase minerals and interfaces.

biotite. Second, the large differences in the elastic modulus between biotite and adjacent minerals (Table 2, Fig. 7) cause strong tensile stress constraints on the crack. Additionally, the angle between the biotite cleavage direction and the crack propagation direction significantly affects the crack propagation behavior, with larger angles resulting in crack deflection or termination. As a result, under the combined effects of these factors, cracks are less likely to penetrate biotite crystals and more likely to propagate along the biotite boundaries. The Qz-Fs interface shows many particulate fragments near the indentation, with a main crack expanding along the interface, exhibiting brittle fracture characteristics. In contrast, the biotite interface displays a more complex failure pattern. The smooth fracture surface and blocky debris around the indentation reflect brittle failure, whereas the strip-like accumulation indicates ductile failure characteristics (Fig. 15d).

3.4. 2D mapping of micro-mechanical properties

A key advantage of grid indentation is generating mechanical property maps for the microscopic minerals of granite. These maps enable the comparison of the relationships between different mechanical properties. Figs. 16–18 display the mechanical property maps for the three minerals. Areas with higher E are often associated with higher H and vice versa. This result demonstrates a strong positive correlation between the two mechanical parameters in all three minerals. These parameters are commonly consid-

Table 3
Macroscopic and microscopic experimental results of fracture toughness.

Reference	Experiment	Type	K_{IC} (MPa·m$^{0.5}$)	σ_K
This study	Nanoindentation	Quartz	2.01	0.19
		Feldspar	1.41	0.20
		Biotite	0.86	0.23
		Qz-Fs	1.63	0.57
		Qz-Bt	1.28	0.48
		Fs-Bt	0.90	0.30
Liu et al. [21]	Nanoindentation	Quartz	5.34	0.07
		Plagioclase	2.70	0.30
		Biotite	2.41	0.34
		Orthoclase	4.83	0.16
		Qz-Bt	2.59	0.46
		Ol-Bt	2.85	0.48
		Qz-Ol	4.96	0.50
		Qz-Pl	4.00	0.76
		Ol-Pl	2.55	0.41
Mahabadi [28]	Nanoindentation	Quartz	8.68	0.18
		Feldspar	4.18	0.09
		Biotite	3.21	0.05
Liu et al. [26]	Notched brazilian disc Splitting	Granite	1.18	
Jeong et al. [29]	Semi-circular bend	Granite	1.06	0.09
	Micro cantilever bend	Granite	0.48	0.35
Wang et al. [27]	Notched three-point-bend	Granite	2.01	0.10

Note: σ_K represents the standard deviation of fracture toughness; Ol the orthoclase; and Pl the plagioclase.

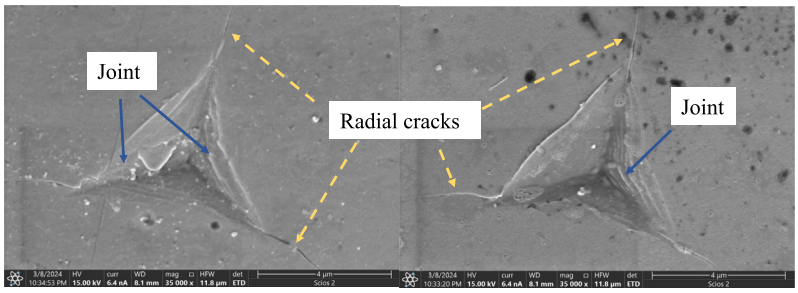
ered to be positively correlated with material strength. However, the relationships between *E*, *H*, and other mechanical properties show distinct differences for different minerals. The distribution of quartz's *P*-value is relatively uniform, and the value does not change substantially with *E* and *H* (Fig. 16d). This finding suggests that quartz's plastic deformation is minimally influenced by its *E* and *H*. In some regions, K_{IC} shows a consistent evolution with the *E* and *H*, although the distribution differs from that of these two mechanical parameters (Fig. 16e). In contrast, feldspar exhibits a notable difference. Regions with a higher *E* and *H* have higher *P*-values (Fig. 17d). However, K_{IC} is relatively high in areas where feldspar's *E* and *H* are lower (Fig. 17e). In comparison, biotite's mechanical properties exhibit a more consistent trend. Its plasticity is lower in regions with higher *E* and *H* and higher in areas with lower *E* and *H* (Fig. 18d). Conversely, K_{IC} is positively correlated with *E* and *H* (Fig. 18e).

In summary, the four mechanical parameters of the three minerals exhibit distinct correlations, reflecting the differences in mechanical properties and their interrelationships across the minerals. Notably, a distinct weak zone is observed in the *E* map of quartz (Fig. 16b). A comparison with Fig. 16a shows that this weakening of *E* is likely due to microcracks on the quartz surface. Additionally, a region of significantly lower K_{IC} is observed around the microcracks. This finding suggests that initial microdefects can reduce the material's resistance to crack initiation. Interestingly, the impact of microcracks is significantly lower on the *H* and *P*-value than on the *E* and K_{IC} (Fig. 16c and d). *H* and *P*-values are strongly correlated with the deformation capacity, suggesting that initial microcracks do not significantly affect the material's deformation. Additionally, the distribution of *P*-values is opposite to that of K_{IC} for quartz, feldspar, and biotite (Fig. 16d and e, Fig. 17d and e, Fig. 18d and e). This result indicates that regions with higher plasticity exhibit stronger crack propagation resistance (Section 3.3) and lower crack initiation resistance.

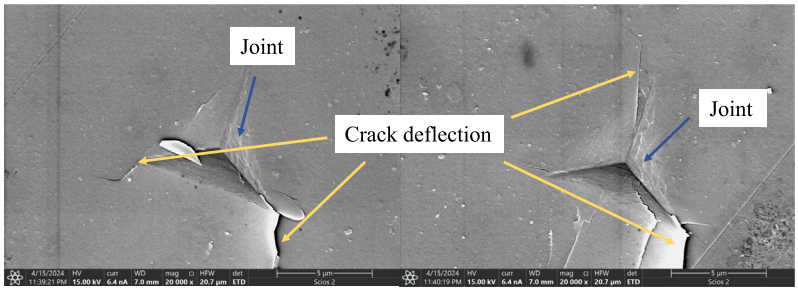
4. Discussion

Nanoindentation involves a multi-physics process encompassing elastic and plastic deformation and ductile and brittle failure of materials. The different characteristics of microscopic components govern the transitions between these physical processes. Therefore, an in-depth investigation of the relationships between nanoindentation mechanical parameters, which control the mechanical behaviors of rocks, deepens our understanding of the interaction mechanisms between these behaviors. Several scholars have discussed the relationships between nanoindentation mechanical parameters. Studies have found a linear relationship between elastic modulus and fracture toughness [32] and a nonlinear relationship with hardness [33]. These findings suggest a high correlation between parameters that characterize different mechanical properties and significant differences in the correlations between different minerals. The results in Section 3.4 corroborate this point. However, the correlations of the mechanical parameters (Section 3.4) are unclear. Therefore, an in-depth investigation of the intrinsic relationships between these mechanical parameters is performed.

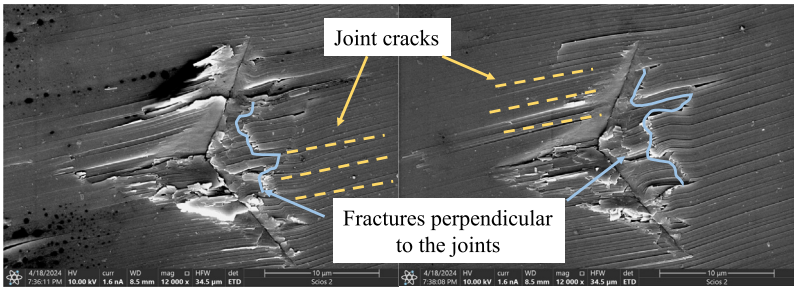
Fig. 19 illustrates the relationship between *E* and *H* for different single-phase minerals. *E* and *H* are generally positively correlated. However, these two mechanical parameters of the three minerals do not show a strong linear relationship. The coefficient of determination (R^2) of feldspar is only 0.49, although it has the best fit. Quartz, with the highest uniformity, has a coefficient of only 0.21. This discrepancy may arise from the higher sensitivity of *H* to microstructural defects than *E*, resulting in higher data variability (Fig. 16b and c). *E* and *H* also do not have a linear relationship for heterogeneous macroscopic aggregates. Liu et al. [21] employed power-law functions and logarithmic functions to characterize the relationship between the *E* and *H* of minerals, respectively. In contrast, the pattern of the *E* and *H* distribution for the mineral aggregates was S-shaped, as illustrated in Fig. 19b. The blue and green fitted curves represent the power-law and logarithmic models, respectively. Although both models exhibit high fitting accuracy (R^2=0.95 and 0.99), their predictive power is limited. This observation is supported by the feldspar data, which predominantly occur on one side of the curve, and the slow increase in *H* with respect to the *E* for biotite and quartz. This trend indicates that these models are inadequate. In contrast, although the S-shaped curve has moderate fitting precision, it provides a more accurate depiction of the relationship between *E* and *H* for the minerals of granite. Additional nanoindentation data from a homologous granite sample



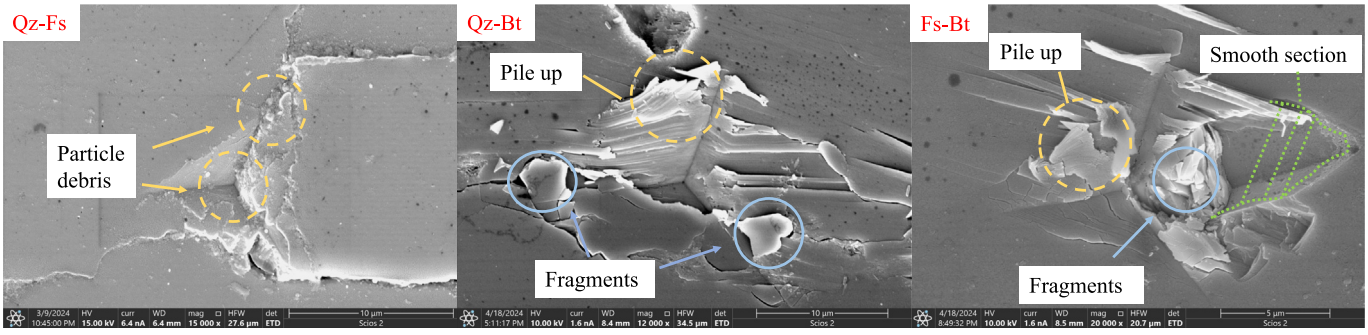
(a) Quartz



(b) Feldspar



(c) Biotite



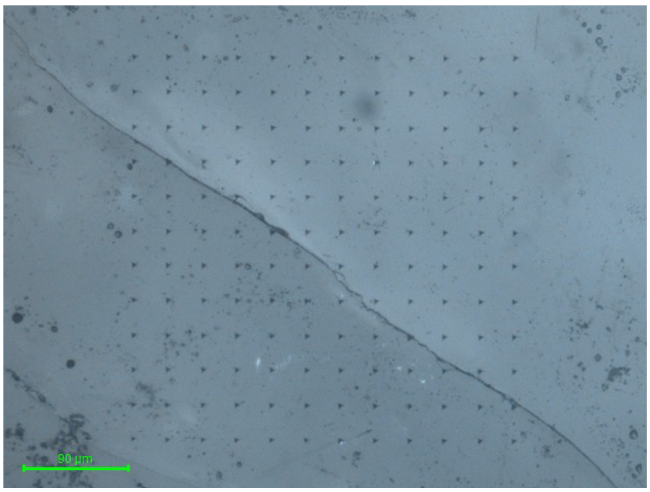
(d) Mineral interfaces

Fig. 15. Fracture characteristics of single-phase minerals and interfaces.

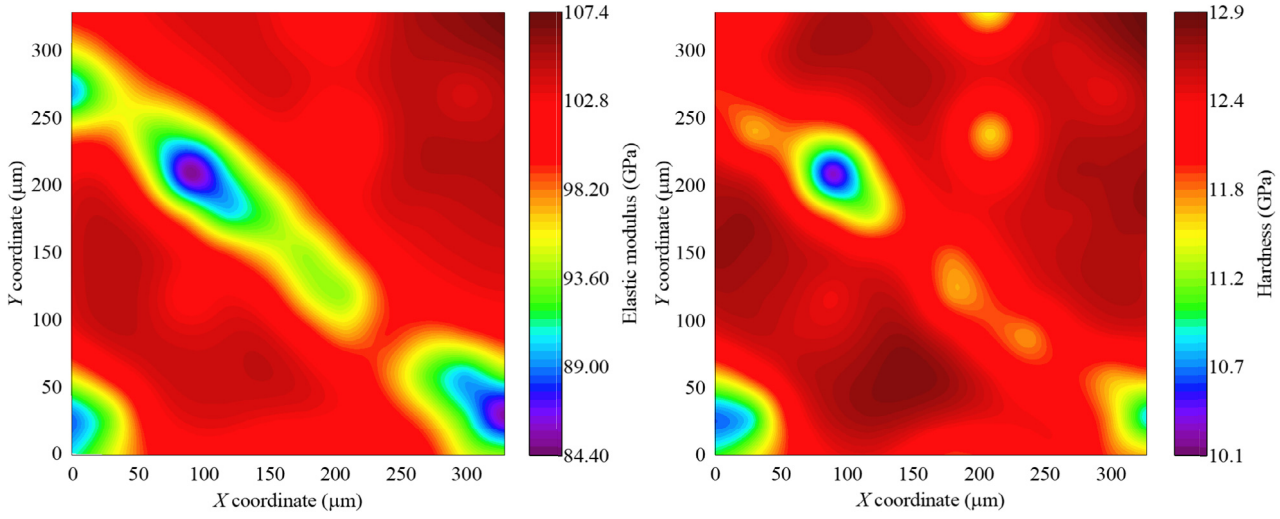
also exhibit a good S-shaped fitting trend (Fig. 19c), further supporting the reliability of this model in characterizing the *E*-*H* evolution in granite.

Fig. 20 illustrates the relationship between *E* and the *P*-value of different single-phase minerals. Similar to the relationship between *E* and *H*, no strong linear correlation is observed between the *E* and the *P*-value of the three minerals. The *P*-value does not exhibit a monotonic relationship with the *E* on the aggregate scale (Fig. 20b). An increase in *E* does not necessarily reduce plastic deformation because the ability of the minerals to undergo plastic deformation is closely related to their inherent structural characteristics (Fig. 12).

Fig. 21 presents the relationship between *E* and K_{IC} of different single-phase minerals. Similarly, no strong linear correlation is observed between these two mechanical parameters for the three minerals. Additionally, no high positive correlation is observed between *E* and K_{IC} of minerals, such as feldspar (Fig. 21a). This may be due to the combined effects of the mineral's heterogeneity, anisotropy, and the high sensitivity of mechanical parameters to microstructural defects, resulting in significant fluctuations in the

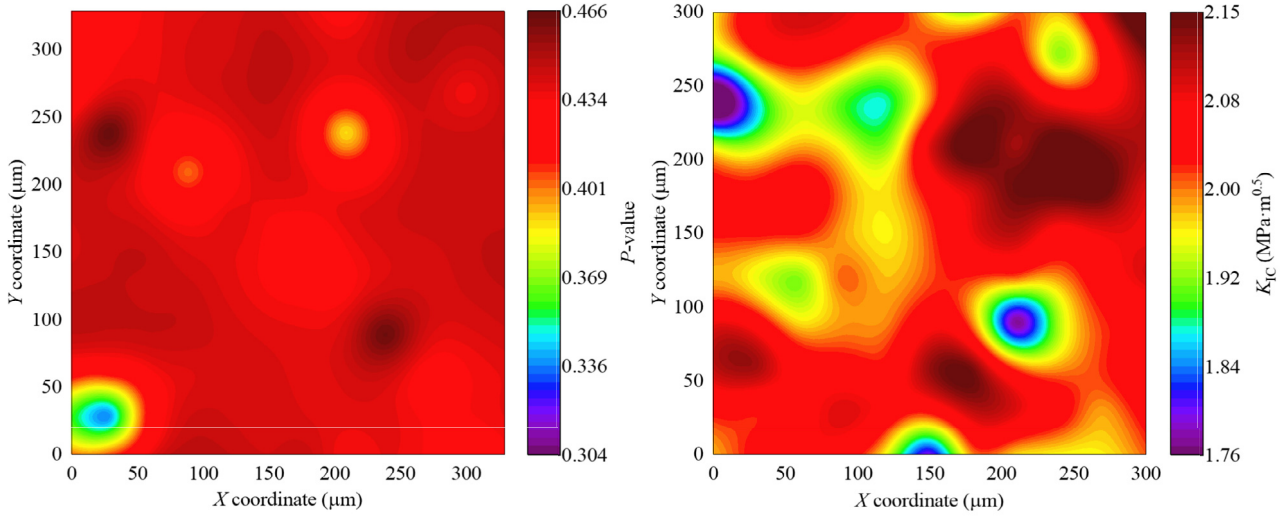


(a) Surface morphology of the grid indentation of quartz



(b) Elastic modulus

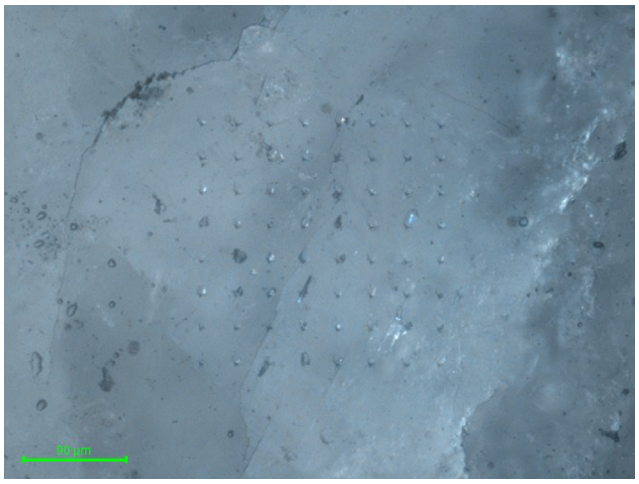
(c) Hardness



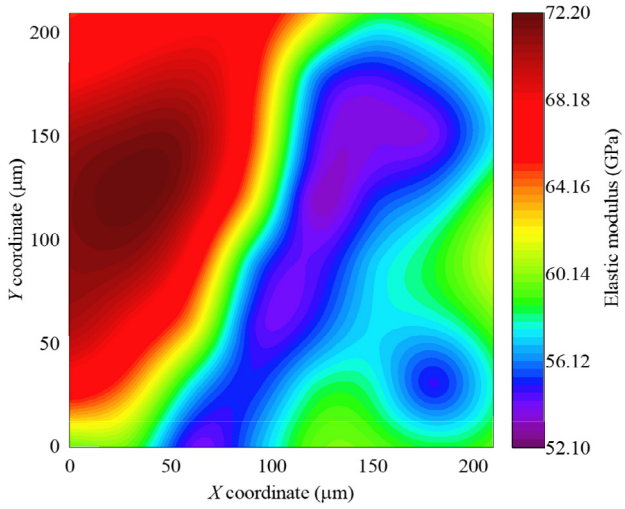
(d) *P*-value

(e) Fracture toughness

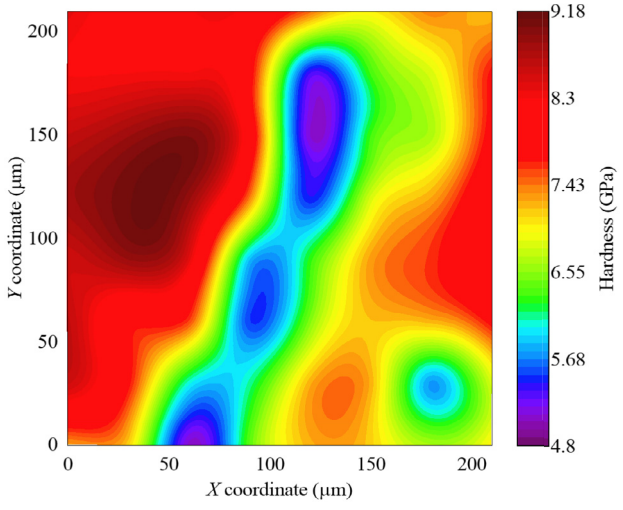
Fig. 16. Nephogram of mechanical properties of quartz.



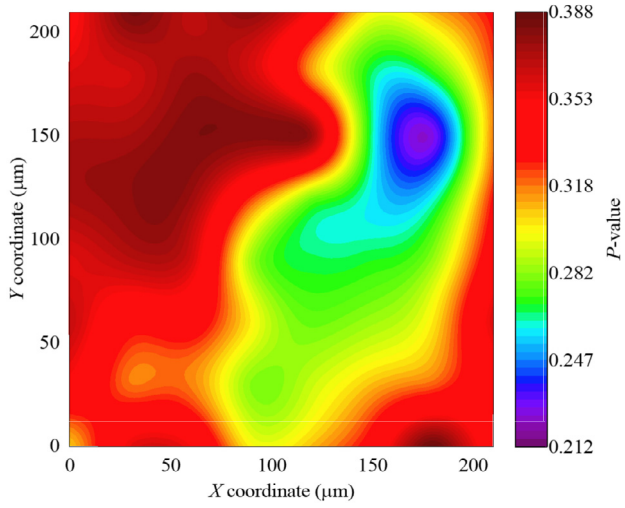
(a) Surface morphology of the grid indentation of feldspar



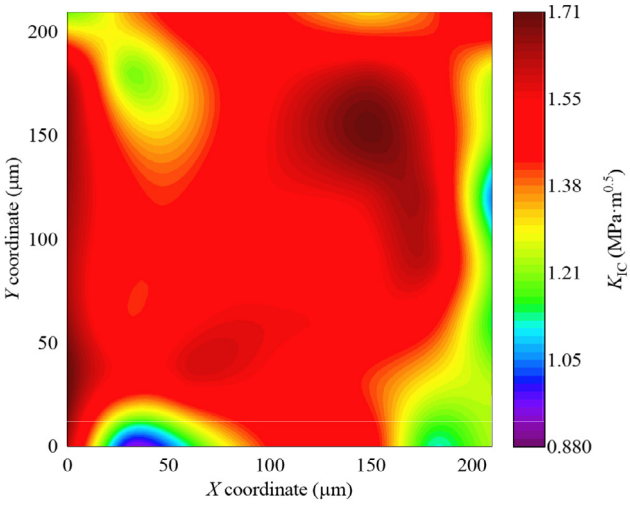
(b) Elastic modulus



(c) Hardness

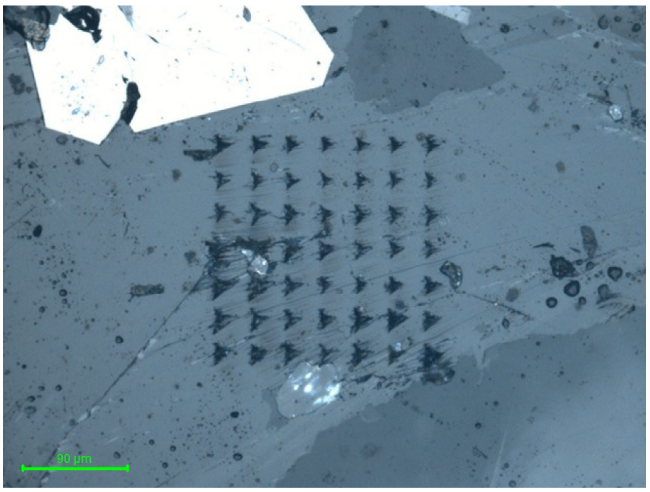


(d) *P*-value



(e) Fracture toughness

Fig. 17. Nephogram of mechanical properties of feldspar.



(a) Surface morphology of the grid indentation of biotite

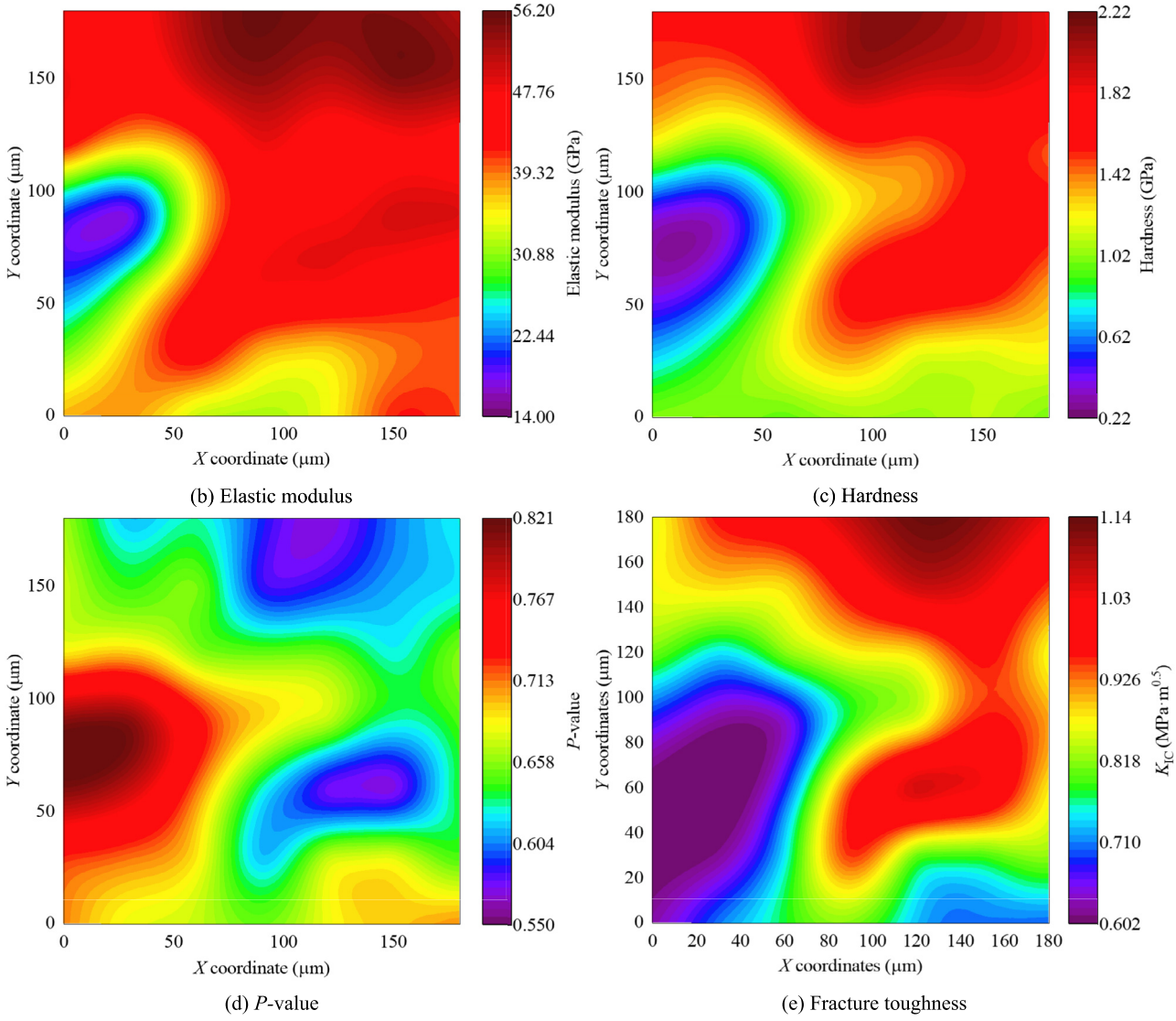


Fig. 18. Nephogram of mechanical properties of biotite.

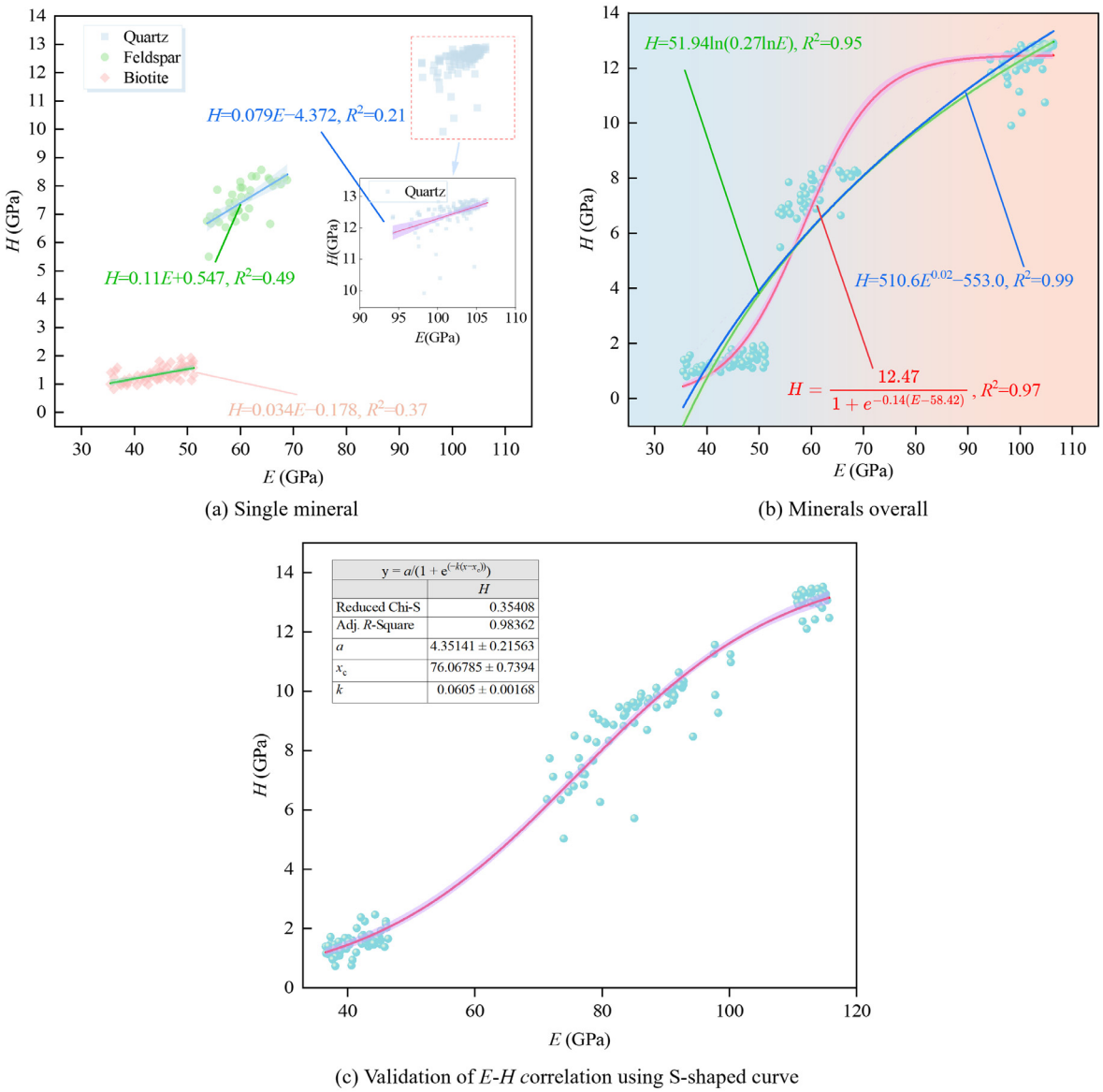


Fig. 19. Relationships between the elastic modulus and hardness.

experimental data. However, a strong positive correlation is observed between E and K_{IC} of mineral aggregates (Fig. 21b), indicating that macroscopic mineral aggregates with a higher E are more resistant to fracture initiation.

In summary, no significant correlation is observed between the different mechanical properties of single-phase minerals, which may be attributed to the different effects of crystallographic anisotropy on their mechanical behavior at microscopic scales. In quartz crystals, the variation in atomic arrangement and bonding direction leads to differences in mechanical responses, with some surfaces being more prone to plastic deformation, while others exhibit greater elastic resistance [34]. Feldspars have a monoclinic crystal system with a higher structural stability than the triclinic system [35], which may lead to a higher modulus. For biotite, the interlayer ionic interactions induce normal anisotropy that predominantly governs its elastic response [36]. These intrinsic anisotropies of the crystal structure enhance the variability in elastoplastic and fracture behavior, which may be the reason for the low correlation among these mechanical parameters. However, at the rock scale, well-defined statistical relationships are observed among mechanical parameters (Figs. 19b, 20b, and 21b). For example, there are significant positive correlations between E and H, and E and K_{IC} (Figs. 19b and 21b). This contrast reveals a distinct scale dependence of the mechanical property relationships in granite. Notably, shale and some other lithologies exhibit positive correlations between the aforementioned mechanical parameters at the rock scale [33,37], but these studies did not systematically examine this scale effect and its intrinsic mechanism. Further research should be carried out in the future.

5. Conclusions

This study examined the microscopic mechanical behavior of Yanshanian granite from the Guangdong-Hong Kong-Macao region. A research framework using SEM-EDS and nanoindentation tests was proposed to investigate the influence of diagenetic minerals and their interfaces on rock deformation and crack propagation. This method can be extended to micromechanical studies of other rocks. The findings also provide valuable experimental results for developing mesoscale numerical models to predict

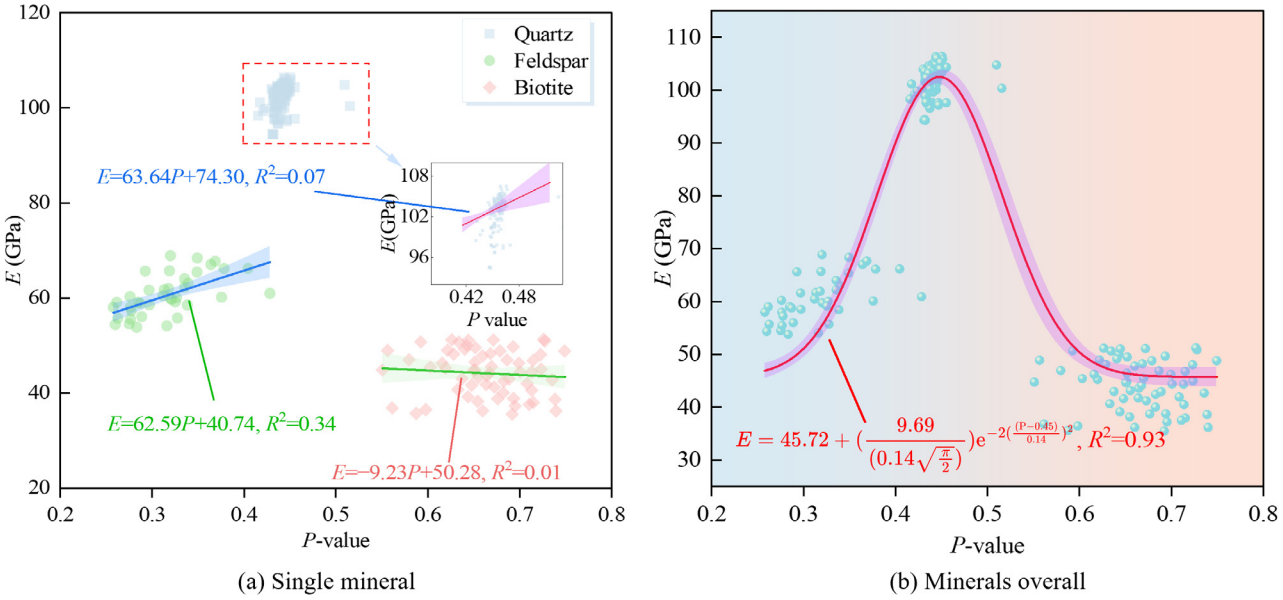


Fig. 20. Relationships between the elastic modulus and plastic index.

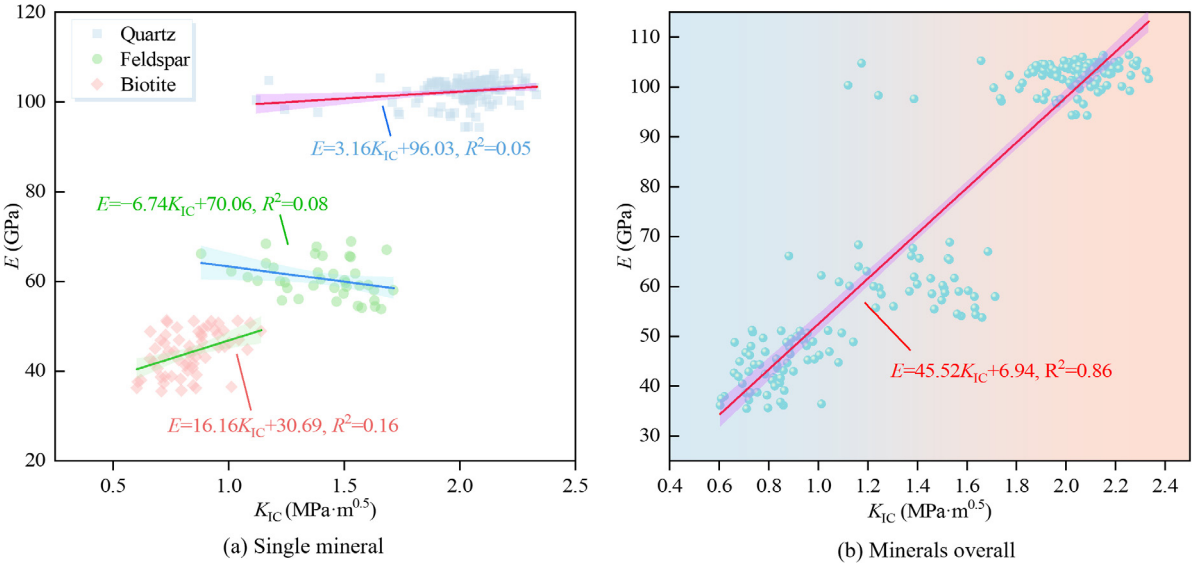


Fig. 21. Relationships between the elastic modulus and fracture toughness.

macroscopic mechanical behavior. The primary conclusions are as follows:

(1) Mineral interfaces generally exhibited mechanical properties falling between those of the two adjoining phases, with greater variability observed in the interfacial properties. According to the magnitude of mechanical properties, the mineral interfaces could be classified into strong interfaces (Qz-Fs) and weak interfaces (biotite interfaces). The E of the microscopic minerals and interfaces were higher than the uniaxial elastic modulus of macroscopic granite, exhibiting a scale effect in mechanical properties.

(2) Biotite and its mineral boundaries exhibited the lowest deformation resistance. The deformed regions were more prone to structural failure. A plasticity index (P-value) was defined to characterize the plastic behavior of minerals and their interfaces. Quartz and feldspar exhibited relatively low P-values, indicative of predominantly elastic deformation characteristics, whereas biotite had the highest P-value, demonstrating pronounced plasticity. Minerals with low P-values exhibited more significant plastic behavior at the interfaces.

(3) The K_{IC} of the minerals and interfaces in granite ranged from 0.86 to 2.02 MPa·m$^{0.5}$, with interfacial K_{IC} values from 0.90 to 1.63 MPa·m$^{0.5}$. The K_{IC} values of the hard minerals, such as quartz and feldspar, were nearly identical to that of the macroscopic granite, indicating that the granite's resistance to crack initiation was primarily attributable to quartz and feldspar. The strong interfaces exhibited brittle fracture behavior, whereas the weak interfaces displayed brittle-ductile fracture characteristics.

(4) Consistent with shale and some other lithologies, significant positive correlations were observed between E and H, and E and K_{IC} in granite at the rock scale. However, no significant

correlation existed between the different mechanical properties of single minerals. The intrinsic anisotropies of mineral's crystal structure increase the variability in elastoplastic and fracture behavior, which may be the reason for the low correlation between elastoplastic and fracture mechanical parameters, causing the scale effect of the mechanical property relationships in granite.

Acknowledgement

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Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijmst.2025.08.009.

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